

GIO - Global Infrastructure Overview

DIPLOMA THESIS

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Leonding, December, 2025

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Abstract

Task

Modern companies operate complex IT infrastructures consisting of numerous servers, services, and resources. Within large infrastructures, information about available systems and their current state is often distributed across multiple platforms. This makes it difficult for employees to maintain an overview and efficiently analyze infrastructure-related data.

As part of this thesis, a custom Grafana plugin for Primetals Technologies Austria GmbH was developed. The objective of the project is the implementation of a Global Infrastructure Overview (GIO), which provides a centralized view of the company's IT infrastructure. Data from various sources is collected, aggregated, and visualized in a user-friendly manner.

Implementation

The implementation was done using a Grafana plugin, divided into multiple layers for maintainability and extensibility. A centralized datasource layer collects and aggregates data from various Grafana data sources, including MSSQL databases and enterprise APIs, before providing it to the visualization layer. Access to the system is secured using Grafana's existing authentication and authorization mechanisms.

The visualization layer displays the collected data on an interactive world map and in detailed views, with filtering and search functionality enabling employees to quickly locate and analyze specific infrastructure information.

Result

A Grafana plugin with the required functionalities was developed, providing a centralized overview of the company's distributed infrastructure through visualizations on a world map and in detailed views. The plugin has already been deployed at Primetals Technologies Austria GmbH, where it is actively used by employees to monitor infrastructure, and can be extended with additional data sources and views.

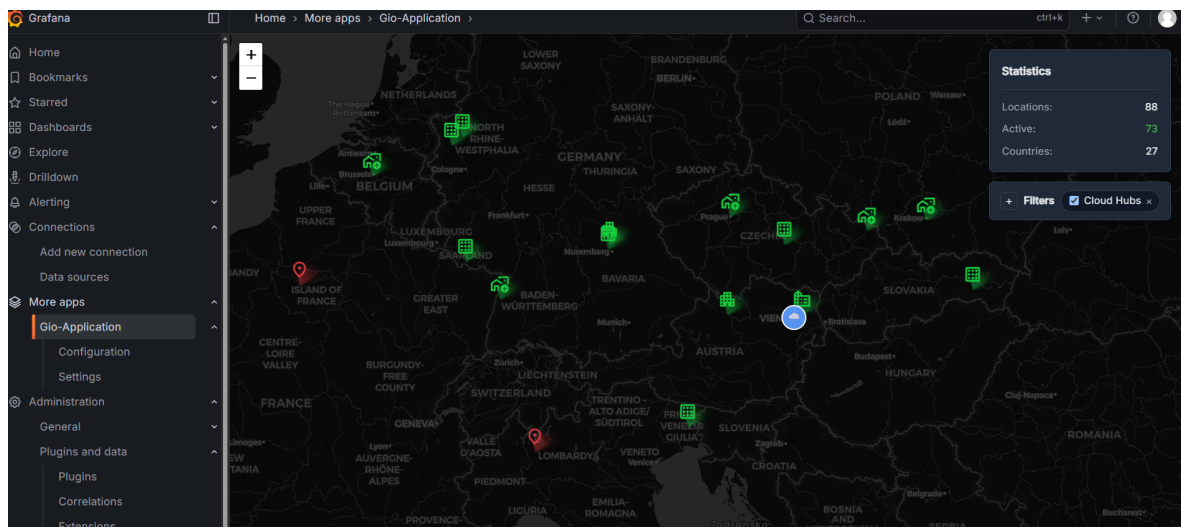


Abbildung 1: Worldmap screenshot

Zusammenfassung

Aufgabenstellung

Moderne Unternehmen betreiben komplexe IT-Infrastrukturen, welche aus einer Vielzahl an Servern, Diensten und weiteren Ressourcen bestehen. Informationen über verfügbare Systeme und deren aktuellen Zustand sind häufig auf mehrere Plattformen verteilt, wodurch eine zentrale Übersicht über die Infrastruktur erschwert wird.

Im Rahmen dieser Arbeit wurde für die Primetals Technologies Austria GmbH ein Grafana-Plugin zur Umsetzung eines Global Infrastructure Overview (GIO) entwickelt. Ziel des Projekts ist die zentrale Darstellung von Infrastrukturinformationen, welche aus unterschiedlichen Datenquellen gesammelt, aggregiert und übersichtlich visualisiert werden.

Realisierung

Die Umsetzung erfolgte durch die Entwicklung eines eigenen Grafana-Plugins, welches zur besseren Wartbarkeit und Erweiterbarkeit in mehrere Schichten unterteilt wurde. Ein zentraler Datasource-Layer sammelt und aggregiert Daten aus verschiedenen Grafana-Datenquellen, darunter MSSQL-Datenbanken und Enterprise-APIs, bevor diese an die Visualisierungskomponente übergeben werden. Der Zugriff auf das System wird über die bestehenden Authentifizierungs- und Autorisierungsmechanismen von Grafana abgesichert.

Die Darstellung der Informationen erfolgt auf einer interaktiven Weltkarte sowie in Detailansichten. Zusätzlich wurden Funktionen zur Suche und Filterung implementiert, wodurch Mitarbeiter gezielt nach Infrastrukturinformationen suchen und diese analysieren können.

Ergebnis

Es wurde ein Grafana-Plugin mit den geforderten Funktionalitäten entwickelt, welches eine zentrale Übersicht über die verteilte IT-Infrastruktur bereitstellt und die Visualisierung aggregierter Daten auf einer Weltkarte sowie in Detailansichten ermöglicht. Die Lösung wird bereits bei der Primetals Technologies Austria GmbH produktiv einge-

setzt und von Mitarbeitern zur Überwachung der Infrastruktur genutzt. Das Plugin kann durch zusätzliche Datenquellen sowie weitere Visualisierungsansichten erweitert werden.

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1 Introduction

1.1 Initial Situation

Employees at Primetals Technologies Austria GmbH currently have to access multiple systems in order to obtain information about the company's IT infrastructure and identify potential issues. According to employees, relevant information is distributed across various sources, primarily MSSQL databases and web interfaces, making infrastructure monitoring time-consuming and inefficient. Employees who have worked with these systems also reported that important information can easily be overlooked due to the lack of a centralized overview.

1.1.1 Company

Primetals Technologies Austria GmbH is an international engineering company specializing in technologies, equipment, and automated solutions for the metals industry. The company develops solutions for steel production with a focus on improving productivity, sustainability, and digitalization [1].

1.1.2 Infrastructure landscape

The infrastructure of Primetals Technologies Austria GmbH consists of multiple sources that contain information about different company resources. The primary MSSQL databases store structured information about devices, locations, and networks through tables and views. Additionally, web applications provide information about devices, Azure services, and the connection status of different networks.

1.2 Problem Statement

The current infrastructure requires employees to manually collect data from independent sources in order to gain a complete understanding of the company's infrastructure. The manual process consumes time and increases the risk of errors during the data analysis. As a result, there is no reliable way to efficiently monitor the current state of the company's IT infrastructure.

1.3 Concept

The proposed concept is based on a centralized application that collects and aggregates information from existing sources. The collected data is processed through a centralized datasource layer, which provides a unified interface for accessing infrastructure information. Instead of manually accessing multiple systems, employees are provided with one centralized visualization that supports them in identifying issues within the company's technical landscape.

1.4 Proposed Solution

The goal is a Grafana plugin that implements the Global Infrastructure Overview (GIO) system. The plugin is divided into multiple layers to separate data processing, communication, and visualization.

The common layer contains repositories and providers, which process and aggregate the collected data before it is accessible through the Grafana Go API interface.

The datasource layer can be extended to retrieve data from MSSQL databases and APIs. The collected information is transformed into a unified structure and forwarded to the visualization layer, which can access the data independently of the original data structure.

The visualization layer presents the data of Primetals Technologies Austria GmbH on a world map and detail views for closer information. The view changes based on zoom and click actions. Additionally, filter and search functions help employees to locate a problem and analyze it.

Access to Grafana is secured through the existing authentication and authorization mechanisms, ensuring that only authorized users can access the plugin.

1.5 Structure of the thesis

The thesis is divided into several chapters. After the introduction, the technical fundamentals of Grafana and the required technologies are presented. The following chapter describes the architecture and implementation of the developed solution. The thesis concludes with an evaluation of the system, discussion of the results and possible future extensions.

2 Glossary

3 Fundamentals of grafana

4 Technologien

4.1 Foo

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4.2.1 Deeper

Nicht mehr im Inhaltsverzeichnis.

Deepest

Vermeide mich.

5 Umsetzung

Siehe tolle Daten in Tab. 1.

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	Regular Customers	Random Customers
Age	20-40	>60
Education	university	high school

Tabelle 1: Ein paar tabellarische Daten

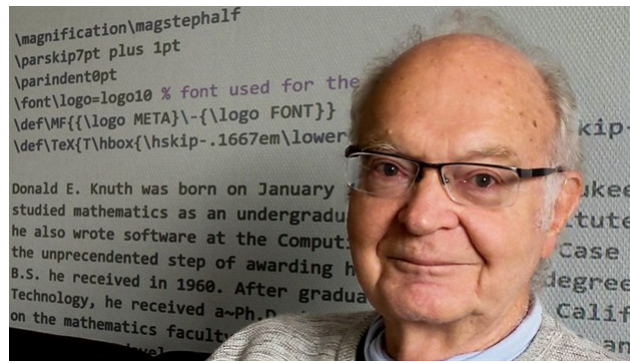


Abbildung 2: Don Knuth – CS Allfather

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Listing 1: Some code

```

1  # Program to find the sum of all numbers stored in a list (the not-Pythonic-way)
2
3  # List of numbers
4  numbers = [6, 5, 3, 8, 4, 2, 5, 4, 11]
5
6  # variable to store the sum
7  sum = 0
8
9  # iterate over the list
10 for val in numbers:
11     sum = sum+val
12
13 print("The sum is", sum)

```

6 Zusammenfassung

Aufzählungen:

- Itemize Level 1
 - Itemize Level 2
 - Itemize Level 3 (vermeiden)
- 1. Enumerate Level 1
 - a. Enumerate Level 2
 - i. Enumerate Level 3 (vermeiden)

Desc Level 1

Desc Level 2 (vermeiden)

Desc Level 3 (vermeiden)

Glossary

Grafana Grafana, an open-source monitoring and visualization software solution that allows data from different sources to be displayed using dashboards and plugins

MSSQL Microsoft SQL Server, a relational database management system used for storing, managing, and retrieving structured data

Literaturverzeichnis

- [1] Primetals Technologies. (2026) Who we are. [Online]. Available: <https://www.primetals.com/en/about-us/who-we-are/>

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